

IE 313 Assignment 3

You need to plan the distribution of goods from a depot located in Manisa, Akhisar to the districts of the Aegean region.

You can use two types of trucks: small truck with a capacity of 10 tonnes and a 10-wheeler with a capacity of 15 tonnes. Small trucks need 20 lt/100 km fuel, and 10-wheelers need 40 lt/100 km. A liter of fuel costs 65 TL. The trucks are located in Akhisar and must return to Akhisar at the end of their tour.

The daily demands (in kg) at each district and real road distances between districts in km are given in the Excel sheet attached.

1. Formulate a capacitated vehicle routing model for this problem as a mixed integer programming problem. Clearly define your parameters, decision variables, objective function and constraints.
2. Implement your model in Pyomo.
3. Solve the problem to optimality using a solver.
4. List your routes and interpret the optimal solution you obtain.
5. Assuming the average speed of both types of trucks is 70 km/hour, and it takes approximately 10 minutes to unload 100 kg from the truck while serving the demand, how long do your tours take?
6. If any of your tours take more than 8 hours, expand your model and introduce a constraint to fit each route in 8 hours. Allow for not satisfying the demand if a shift of 8 hours is not sufficient. Resolve the model. How does your solution change? Report unsatisfied demand if any.